Problem

Consider a graph G with edges E and vertices V, and subgraphs $G_1, \ldots, G_k \subset G$. Let $G_i = (E_i, V_i)$ for these subgraphs. For now let us suppose the graph is unweighted and undirected. The subgraphs need not cover G and can overlap. Each player i knows G and G_i, but not the other player's graphs. (This is akin to a map of a city, where each player is a different regional taxi company.) The goal is, given some starting vertex $v \in V$, and some end vertex, $w \in V$ to:

1. Identify if it is possible to construct a path $v, v_1, \ldots, v_l, w$ such that every edge on the path is in $\bigcup_{i=0}^{k} E_i$.
2. If such a path exists, find the minimum number of edges needed to get from v to w, while using only edges belonging to $\bigcup_{i=0}^{k} E_i$.

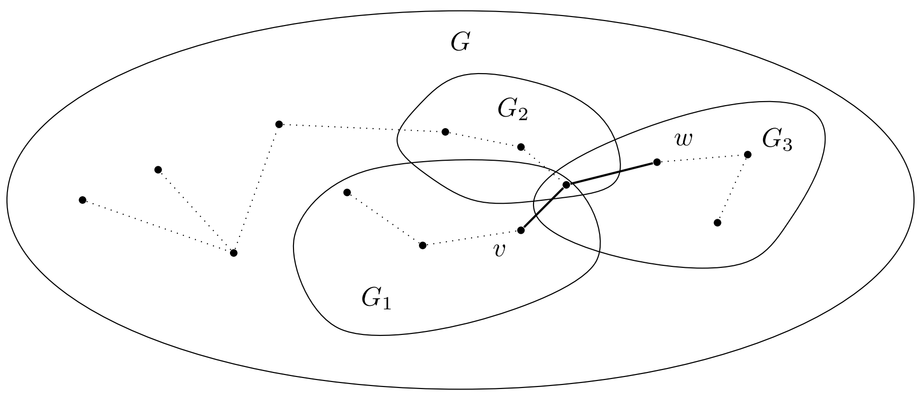

Figure 1: A valid path from v to w in bold. Note that G may contain nodes not belonging to any player.

Lower bound sketches and ideas

For item 1, we think this would require computing $\binom{k}{2}$ pairwise set disjointnes in the worst case, to determine where the intersections of each G_i are, at which point each player could on their own determine the possible paths between each of the subgraphs that intersect with their own.

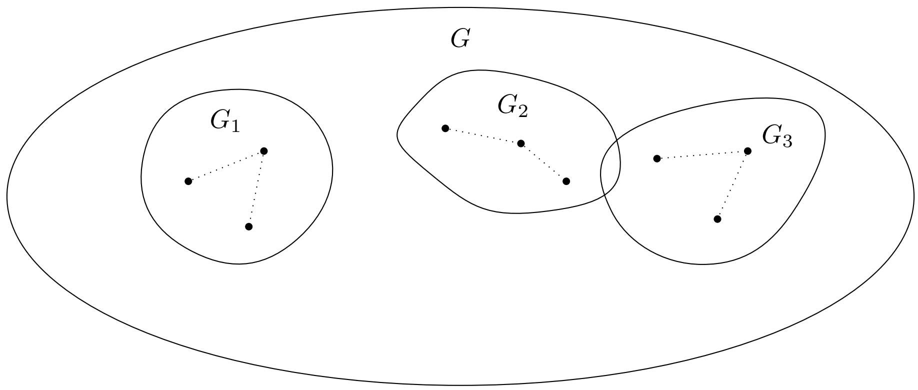

Figure 2: Worst case for item 1: Disjoint Subsets

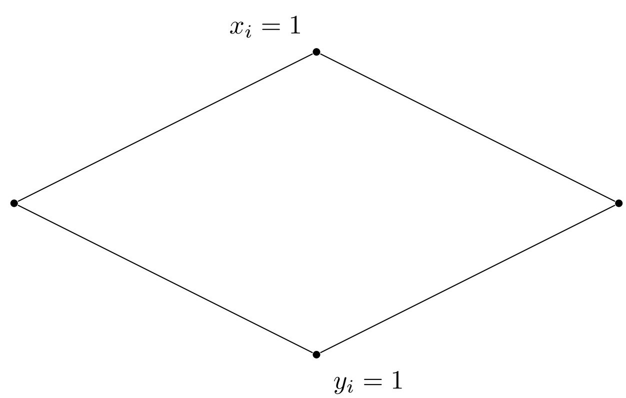

Figure 3: An example of the gadget for set disjointness, where each gadget can be encoded as the compliment for the set union

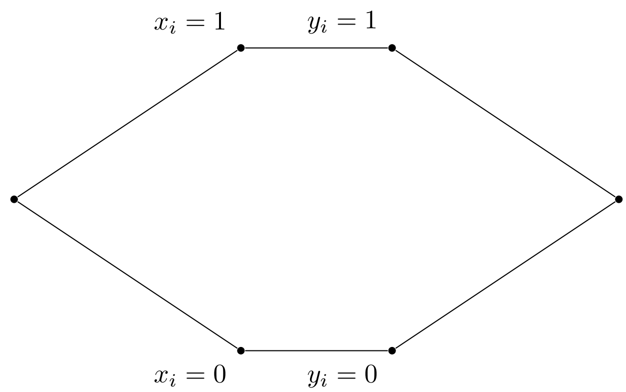

Figure 4: Gadget for proving equality. A path only exists if $x_i = y_i$ for either branch

For item 2, we would be required to compute $\binom{k}{2}$ pairwise intersections, since in order to minimize the path we would need to know all possible paths that could connect v and w. To find the shortest path, each player works backwards and sends the distances through their internal vertices to the players they intersect with, and each player will attempt to identify if a longer path can be dropped whenever they receive this list.